CHAPTER 56 | AGENT FACTORY | OPENCLAW

# Meet Your Personal AI Employee

## Complete Chapter Notes — Sab 15 Lessons

Roman Urdu Study Notes | PDF Ready

**Prepared for: Imran Hassan | 2026**

Source: agentfactory.panaversity.org

Chapter 56 — Building OpenClaw Apps

**15 Lessons | Complete Coverage**

# 📋 Chapter 56 — Lesson Index


1 The AI Employee Moment — 6 Dimensions

2 Install & Connect Your Employee

3 Delegate Real Work — Agent Loop

4 Customize Your Employee's Brain

5 Memory & Commands — MEMORY.md

6 Install Skills & Discover Ecosystem

7 Connect External Tools — MCP Servers

8 Make It Proactive — Heartbeats & Cron

9 Give It a Voice — TTS

10 Add a Second Agent

11 Connecting Google Workspace

12 Orchestrate Other Agents

13 Gate Your Agent's Tools — Plugins

14 Deploy to Production — VPS

15 Isolate with NemoClaw — 4-Pod Security

# 1 The AI Employee Moment

Concept | 6 Dimensions | Agent OS

🎯 **IS LESSON MEIN KYA SIKHENGE**

Personal AI Employee aur aam chatbot mein kya farq hai — 6 dimensions ke zariye samajhna. Agent OS analogy se OpenClaw ka architecture samajhna.

🍴 **RESTAURANT VS PERSONAL COOK**

**Chatbot = Restaurant:** Tab kholo → prompt likho → jawab lo → band karo. Agle waqt zero se shuru. Agent aapke ghar nahi aata, fridge nahi check karta, kuch yaad nahi rakhta.

**AI Employee = Personal Cook:** Ghar mein rehti hai, fridge khud check karti hai, instructions kahin se bhi do (call, SMS, note), nayi recipes seekh sakti hai, dusri cook bhi kaam kar sakti hai, sab data aapke ghar mein.

## 6 Dimensions — Kya Banata Hai AI Employee Ko?

| # | Dimension | Chatbot | AI Employee |
|---|---|---|---|
| 1 | **Multi-Channel** | Sirf ek platform | WhatsApp, Telegram, Discord, Slack — sab pe ek saath, ek hi memory |
| 2 | **Always-On** | Tab kaam karta hai jab tab khulta hai | Background daemon — 24/7 silently running |
| 3 | **Proactive** | Sirf tab kaam karta hai jab aap type karo | Heartbeats (har 30 min) + Cron jobs — khud check karta hai |
| 4 | **Extensible** | Jo features company de, bas wahi | Plugins, Skills, MCP servers — nayi capabilities add kar sakte ho |
| 5 | **Multi-Agent** | Ek AI sab kuch karta hai | Multiple specialists — ek gateway ke peeche alag alag agents |
| 6 | **Ownership** | Aapka data unke server pe | Aapke machine pe — aapka data, aapki files, aapki marzi |

🔑 **Sabse Important Dimension: Ownership**

Baaki 5 dimensions ise employee banate hain. Ownership ise AAPKA employee banata hai. Salesforce jaise enterprise tools mein bhi pehli 5 hoti hain — lekin vendor control karta hai sab kuch.

# 2 Install & Connect Your Employee

Hands-On | Install | WhatsApp/Telegram/Discord | Troubleshooting

🎯 **IS LESSON MEIN KYA SIKHENGE**

OpenClaw install karna, sabse common installation failure se bachna, aur pehla messaging channel connect karna. Universal setup pattern seekhna.

## Install Commands

**macOS / Linux:**

```
curl -fsSL https://openclaw.ai/install.sh | bash
```

**Windows PowerShell (Admin):**

```
iwr -useb https://openclaw.ai/insta
```

## WSL Setup (Windows ke liye Recommended):

```
wsl --install -d Ubuntu          # Step 1: Ubuntu install
curl -fsSL https://openclaw.ai/install.sh | bash  # Step 2: OpenClaw
openclaw onboard --install-daemon  # Step 3: Wizard + daemon
```

## LLM Provider — Google Gemini (Free)

| Model | Req/Min | Req/Day | Recommendation |
|---|---|---|---|
| gemini-3.1-flash-lite-preview | 15 | 500 | ✅ Best for learning |
| gemini-2.5-flash | 10 | 250 | Backup |
| gemini-2.5-pro | 5 | 50 | Heavy tasks |

## Troubleshooting — 3 Common Problems

⚠️ **1. Crash Loop — gateway.mode missing**

Fix: `openclaw config set gateway.mode local` → `openclaw gateway restart`

⚠️ **2. Auth Cache Gotcha**

API key rotate karne ke baad cached key use hoti hai. Fix: `rm ~/.openclaw/agents/main/agent/auth-profiles.json`

⚠️ **3. No Response (30 sec)**

Fix: `openclaw channels status --probe` → `openclaw doctor` → `tail -f ~/.openclaw/logs/gateway.log`

⭐ **UNIVERSAL SETUP PATTERN (5 STEPS)**

1) Runtime install → 2) Intelligence configure (LLM) → 3) I/O connect (channel) → 4) End-to-end verify → 5) Security secure (pairing mode)

# 3 Delegate Real Work

Agent Loop | Tool Profiles | Dashboard

🎯 **IS LESSON MEIN KYA SIKHENGE**

Agent loop ke 5 steps trace karna. Tool badges samajhna (knowledge vs machine vs internet). Tool profiles — kya allowed hai, kya nahi.

## 3 Tasks — 3 Alag Paths

| Task | Query | Kya Chahiye | Dashboard Badge |
|---|---|---|---|
| A | "Tell me about yourself" | Knowledge (training data) | ❌ Koi badge nahi |
| B | "List 5 largest files in home dir" | Aapki machine | ✅ "Exec" badge |
| C | "Search web for AI Agent Factory book" | Internet (live search) | ✅ "web_search" badge |

## Agent Loop — 5 Steps

1. **INTAKE:** Gateway message receive aur validate karta hai
2. **CONTEXT ASSEMBLY:** Workspace files, skills, bootstrap prompt build karta hai
3. **MODEL INFERENCE:** LLM decide karta hai kya karna hai, kaun sa tool chahiye
4. **TOOL EXECUTION:** Agar model ne tool choose kiya to run hota hai
5. **REPLY & PERSIST:** Response stream hota hai aur session history mein save hota hai

## Tool Profiles

| Profile | Kya Include Hai | Kya Disabled Hai |
|---|---|---|
| coding (default) | File I/O, exec, sessions, memory, image | — |
| messaging | Sirf messaging tools | File access, exec, web search |
| minimal | session_status only | Almost sab kuch |

| | | |
|---|---|---|
| full | Sab tools, unrestricted | Kuch nahi |

Profile change karo Dashboard → Agents → Tools → Quick Presets. Koi restart nahi chahiye — immediately apply hota hai.

# 4 Customize Your Employee's Brain

Workspace Files | SOUL.md | IDENTITY.md | Emma's Rule

🎯 **IS LESSON MEIN KYA SIKHENGE**

Workspace files edit karke agent ki personality real-time mein badalna. Emma's rule: agar instruction base model se alag behavior nahi deti — delete karo.

## Workspace Files — ~/.openclaw/workspace/

| File | Purpose | Kab Load Hoti Hai |
|---|---|---|
| `SOUL.md` | Persona, tone, boundaries | Har session, har message |
| `IDENTITY.md` | Agent ka naam, emoji, vibe | Har session |
| `AGENTS.md` | Operating instructions, rules, priorities (employee handbook) | Har session |
| `TOOLS.md` | Local tools ki guidance notes (tool access nahi deta) | Har session |
| `USER.md` | Aapka naam, address preference, timezone | Har session |
| `HEARTBEAT.md` | Periodic self-check checklist | Har 30 minutes |
| `MEMORY.md` | Long-term curated memory | Har session (sirf main private) |
| `memory/YYYY-MM-DD.md` | Daily journal logs | Aaj + kal automatically |

⚠️ **TOOLS.md Misconception:**

TOOLS.md tool access nahi deta — sirf guidance notes hain. Tool access tool profiles se control hota hai (Lesson 3).

💡 **Emma's Rule:**

Har workspace instruction ke baare mein pucho: "Kya agent bina is line ke alag behave karta?" Agar jawab "nahi" hai — delete karo. Base model pehle se helpful hai — use dobara kehna context waste hai.

## Agent Se Apna SOUL.md Edit Karwao:

```
Edit your SOUL.md. Remove any instruction you would follow by default.
Keep only rules that make you different from the base model.
Be direct and concise. Match response length to question complexity.
```

✅ **Community Personality Upgrade Prompt:**

Agent ko bhejo: opinions rakho, corporate rules delete karo, "Great question" se kabhi shuru mat karo, brevity mandatory hai, humor allowed hai, dumb kaam karo to seedha bolo. Phir test karo.

# 5 Memory & Commands

MEMORY.md | memory/ Directory | Slash Commands

🎯 **IS LESSON MEIN KYA SIKHENGE**

Preference store karna jo sessions ke baad bhi rahe. MEMORY.md aur daily logs ka farq. Slash commands — zero tokens, instant response.

## Memory System — 2 Locations

| Location | Kya Store Hota Hai | Kab Load Hoti Hai |
|---|---|---|
| `MEMORY.md` | Curated long-term memory: preferences, facts, decisions | Har session start pe |
| `memory/YYYY-MM-DD.md` | Daily journal: session notes, conversation summaries | Aaj + kal automatically; baaki memory_search se |

⚠️ **Verbal Confirmation Enough Nahi Hai!**

"Got it, noted!" — ye proof nahi ke file write hua. Dashboard mein tool badge check karo. Agar badge nahi to file nahi lika. "Save in memory:" se start karo request.

## Memory Save Karo:

```
Save in memory: I prefer all summaries under 100 words as bullet points.
Never use numbered lists for summaries.
```

## Verify On Disk:

```
cat ~/.openclaw/workspace/MEMORY.md
```

## Slash Commands — Zero Tokens!

| Command | Kya Karta Hai |
|---|---|
| `/help` | Available commands list |
| `/status` | Model, session state, quota info |
| `/model <name>` | Mid-conversation model switch |
| `/reset` | Fresh session (memory safe) |

| | |
|---|---|
| `/compact` | Older turns summarize karo, context free karo |
| `/commands` | Sab available commands (plugins ke baad update hota hai) |

✅ **Slash vs Natural Language:**

/status → instant, no thinking indicator, no token spend. "What model are you using?" → slow, model process karta hai, tokens waste. Slash commands gateway intercept karta hai — model tak pahuncha hi nahi.

# 6 Install Skills & Discover Ecosystem

Skills | MCP | Plugins | ClawHub | Escalation Path

🎯 **IS LESSON MEIN KYA SIKHENGE**

3 extension types: Skills, MCP servers, Plugins. ClawHub se skill install karna. Escalation path: Skills → MCP → Plugin.

## 3 Extension Types

| Type | Kya Hai | Kab Use Karein | Format |
|---|---|---|---|
| **Skill** | SKILL.md folder — knowledge + workflow | Domain knowledge, conversational patterns, decision frameworks | SKILL.md + scripts/ + references/ |
| **MCP Server** | External service access via protocol | Live data, APIs, external systems (e.g., time, database) | JSON config + server process |
| **Plugin** | Gateway capability change | Gateway-level features (TTS, hooks, approval gates) | TypeScript package |

⭐ **ESCALATION PATH**

Pehle Skill try karo → agar external access chahiye to MCP → agar gateway capability chahiye to Plugin. Plugin last resort hai.

## ClawHub — Skill Marketplace

```
openclaw skills search booking       # Search karo
openclaw skills install service-booking  # Install karo
openclaw gateway restart              # Restart zaroori hai!
openclaw skills list                  # Verify karo
openclaw skills update --all          # Sab update karo
```

⚠️ **Gateway Restart Zaroori Hai!**

Skills session start pe snapshot hoti hain. Install ke baad restart karo warna naya skill kaam nahi karega.

## SKILL.md Structure:

```
skill-name/
├── SKILL.md          # Required: metadata + instructions (YAML frontmatter)
```

```
├── scripts/          # Optional: executable code (Python, Bash, JS)
└── references/       # Optional: documentation
```

💡 **Progressive Disclosure:**

Sirf naam aur description startup pe load hote hain (token light). Full instructions sirf tab load hoti hain jab skill activate ho. Cross-platform: OpenClaw, Claude Code, dusre platforms mein same SKILL.md format kaam karta hai.

# 7 Connect External Tools

MCP Server | Silent Failure | Transports | Gateway Log

🎯 **IS LESSON MEIN KYA SIKHENGE**

Pehla MCP server connect karna. OpenClaw ka sabse counterintuitive behavior: silent failure — chat mein koi error nahi, sirf log mein. 3 transport options.

## Time Server Connect (Example):

```
openclaw config set mcp.servers.time '{"command":"uvx","args":["mcp-server-time"]}'
openclaw gateway restart
```

## Verify:

```
openclaw mcp list        # Sab servers
openclaw mcp show time   # Ek server ki details
```

## MCP Commands:

```
openclaw mcp set remote-tools '{"url":"https://mcp.example.com"}'
openclaw mcp unset time
```

## 3 Transport Types:

| Server Type | Transport | Config |
|---|---|---|
| Local (npm/Python script) | stdio | command + args |
| Remote (HTTP SSE) | SSE | url (default for URL) |
| Remote (HTTP streaming) | streamable-http | url + "transport":"streamable-http" |

⚠️ **Silent Failure — Sabse Dangerous Bug!**

Galat package naam → server start nahi hota → agent chat mein koi error nahi dikhata → training data se jawab deta hai. Log check karo: `tail -20 ~/.openclaw/logs/gateway.log`

| Common Cause | Log Mein Kya Dikhta Hai | Fix |
|---|---|---|
| Wrong package name | Non-zero exit code | Package naam verify karo |

| | | |
|---|---|---|
| Missing runtime | "uvx: command not found" | uv ya Node.js install karo |
| Remote unreachable | Connection timeout | URL check karo |
| Gateway not restarted | Old config active | openclaw gateway restart |

💡 **Key Insight:**

MCP servers additive hain — har config set command nayi tools add karta hai bina purani tools affect kiye. Gateway config sab agents ke liye shared hai.

# 8 Make It Proactive

Heartbeats | Cron Jobs | HEARTBEAT_OK | Cost Math

🎯 **IS LESSON MEIN KYA SIKHENGE**

Heartbeats aur Cron jobs — agent khud kaam kare bina aapke puche. HEARTBEAT_OK suppression. Cost math: 1 heartbeat 5 crons se 80% sasta.

🏭 **WAREHOUSE ANALOGY**

**James ki baat:** Floor supervisor har ghante ek baar sab aisles check karta tha. Ek banda, 12 aisles, ek trip. Kuch thik tha to koi call nahi. Pallet galat tha to radio call. Yahi Heartbeat hai.

**Cron = Exact time delivery:** "Roz 9 baje daily summary bhejo" — heartbeat se alag hai.

## Heartbeat vs Cron — Cost Math

❌ **5 Separate Crons (Galat)**

5 × 30 min × 24 hours = 240 agent turns/day

✅ **1 Heartbeat + 1 Cron (Sahi)**

48 heartbeat turns + 2 cron turns = 50/day → 80% reduction!

## Heartbeat Setup:

```
# HEARTBEAT.md mein items add karo (agent se)
# Phir routing set karo:
openclaw config set agents.defaults.heartbeat.every 1m   # Test ke liye
openclaw config set agents.defaults.heartbeat.target last  # WhatsApp pe
openclaw gateway restart
```

## HEARTBEAT_OK — Silent Mode:

💡 **HEARTBEAT_OK Token:**

Agent checklist check karta hai → agar kuch attention nahi chahiye → sirf "HEARTBEAT_OK" token return karta hai → gateway message drop karta hai silently. Aapke phone pe kuch nahi aata!

```
# Daily use ke liye:
openclaw config set agents.defaults.heartbeat.every 30m
# Specific number pe bhejne ke liye:
```

```
openclaw config set agents.defaults.heartbeat.target whatsapp
openclaw config set agents.defaults.heartbeat.to '"+923001234567"'
```

### Design Principle:

Agent control karta hai **WHAT** check karna hai (HEARTBEAT.md). Operator control karta hai **HOW OFTEN** aur **WHERE** (gateway config).

# 9 Give It a Voice

TTS | 4 Modes | Microsoft Edge | OpenAI | Activation Dance

🎯 **IS LESSON MEIN KYA SIKHENGE**

TTS enable karna, 4 auto modes samajhna, aur janna ke hard part speaking nahi — balke KAB bolna ye decide karna hai.

## 4 TTS Providers:

| Provider | Key Chahiye? | Quality | Cost |
|---|---|---|---|
| Microsoft Edge | ❌ Nahi | Good | Free — 300+ neural voices |
| OpenAI TTS | ✅ Haan | Excellent | ~$15/M chars |
| ElevenLabs | ✅ Haan | Premium | Free tier available |
| MiniMax | ✅ Haan | Good | Free tier available |

## Enable (Microsoft — No API Key):

```
openclaw config set messages.tts.auto always
openclaw config set messages.tts.provider microsoft
openclaw config set plugins.entries.microsoft.enabled true
openclaw gateway restart
```

## 4 TTS Modes:

| Mode | Kaun Decide Karta Hai | Behavior | Use Case |
|---|---|---|---|
| `off` | Nobody | Sirf text | Default |
| `always` | Config | Har reply voice note | Pipeline test only — annoying! |
| `inbound` | Customer | Voice to voice, text to text | ✅ Production best choice |
| `tagged` | Agent | [[tts]] tag pe voice | Theory mein flexible, practice mein buggy |

## Inbound Mode (Recommended):

```
openclaw config set messages.tts.auto inbound
# No gateway restart needed — dynamic apply hota hai
```

✅ **Activation Dance yaad rakho (Lesson 2 se):**

1) Plugin list check → 2) Default disabled → 3) Enable → 4) Configure. Har feature isi pattern pe kaam karta hai!

💡 **1500 char limit:**

Replies longer than 1500 chars auto-summarize ya skip ho jaati hain. Adjust: `/tts limit 3000` on WhatsApp. Check: `/tts status`

# 10 Add a Second Agent

Multiple Agents | Routing | Workspace vs Gateway Config

🎯 **IS LESSON MEIN KYA SIKHENGE**

Same WhatsApp pe doosra agent create karna. Routing rules se alag alag agents ko messages route karna. Gateway vs workspace config ka farq.

🏭 **WAREHOUSE ANALOGY**

James ka warehouse: ek desk pe sab — drivers, customers, suppliers. Solution: desk split karo. Inbound shipments left, customer pickups right. Different queues, different permissions. Yahi multi-agent hai!

## Second Agent Create Karo:

```
openclaw agents add helper-agent       # Create
openclaw agents list --bindings         # Verify
openclaw agents bind --agent helper-agent --bind whatsapp  # Bind
openclaw gateway restart
```

## Gateway vs Workspace Config:

| Config Type | Scope | Examples |
|---|---|---|
| Gateway Config | Sab agents ke liye shared | Plugins, TTS, Model |
| Workspace Config | Per-agent (individual) | SOUL.md, skills, MCP |

⚠️ **Session Cache — Changes Immediately Apply Nahi Hote!**

Identity changes next session pe effect karte hain, active conversation mein nahi. Production mein: customers active conversations mein purana persona dekhte rahenge.

## Routing Rules:

```
# Sender-specific routing
openclaw agents route --agent helper-agent --from "+923001234567"
# Unbind
openclaw agents unbind --agent helper-agent --bind whatsapp
```

💡 **Skills Isolation:**

Ek agent ke workspace ki skills doosre agent ko available nahi hoti. Intentional design: customer-facing agent admin-only skills nahi dekhna chahiye.

# 11 Connecting Google Workspace

gog CLI | Gmail | Calendar | Drive | OAuth | Least Privilege

🎯 **IS LESSON MEIN KYA SIKHENGE**

gog CLI se Gmail, Calendar, Drive connect karna. OAuth scope limit karna (least privilege). Secrets audit. Ye pehla waqt hai jab agent aapka REAL data dekhega.

⚠️ **Test Account Use Karo!**

Apna primary Google account use mat karo — ek dedicated test account se shuru karo. Agent ko email access milne ke baad password resets, financial statements sab dikh sakta hai.

## gog — Kya Connect Karta Hai:

| Service | Agent Kya Kar Sakta Hai |
|---|---|
| Gmail | Search, read, send, manage labels, drafts |
| Calendar | List events, create meetings, check availability, respond to invitations |
| Drive | Search files, upload/download, manage permissions |
| + aur | Contacts, Sheets, Docs, Slides, Forms, Chat, Tasks, Keep, etc. |

## Install gog:

```
# macOS
brew install gogcli
# Windows/Linux
npm install -g gogcli
# Verify
gog --version
```

## Setup Steps:

1. console.cloud.google.com → Project create karo
2. APIs & Services → Library → Gmail, Calendar, Drive APIs enable karo

3 OAuth 2.0 credentials create karo (Desktop app type)

4 `gog auth login` → browser se Google login

5 `openclaw secrets audit` → credentials safely stored hain?

```
gog auth services       # Available services list
gog gmail --json        # Gmail JSON output (OpenClaw parse karta hai)
gog calendar            # Calendar commands
gog drive               # Drive commands
```

# 12 Orchestrate Other Agents

sessions_spawn | ACP | Model Limitation | Concurrency

🎯 **IS LESSON MEIN KYA SIKHENGE**

Agents ko ek doosre pe delegate karna. sessions_spawn (blocking) vs ACP spawn (async). Model limitation — free-tier models tools ignore karte hain.

⚠️ **Model Limitation — Sabse Pehle Jaano!**

Free-tier models sessions_spawn tool dekhte hain lekin use nahi karte — hallucinated reasons dete hain ("I don't have permission", "limited to specific sessions"). Ye platform bug nahi — model ki weakness hai. Workaround: session reset karo aur direct aur persistent raho.

## sessions_spawn — Subagent Spawn Karo:

```
Do you have the sessions_spawn tool? Use it now. Spawn a subagent
that researches [your topic]. Do not answer directly.
```

## Subagent Management:

```
/subagents list    # Active aur recent subagents
/subagents kill #1 # Ek subagent stop karo
```

## Concurrency Model:

**Session Lane (per customer)**

Har customer ka apna isolated queue hota hai

**Global Lane (shared)**

Sab agents share karte hain — bandwidth limit

## Nested Orchestration Enable Karo:

```
openclaw config set agents.main.maxSpawnDepth 2
# Default: 0 (subagents khud spawn nahi kar sakte)
```

✅ **Production ke liye:**

Claude Sonnet ya GPT-4 class model use karo — automatically delegate karta hai bina push ke. $50-100/month model cost production orchestration ke liye optional nahi hai.

# 13 Gate Your Agent's Tools

3-Tier Safety | requireApproval Plugin | Spec-Driven Build

🎯 **IS LESSON MEIN KYA SIKHENGE**

Approval plugin banana — agent se koi shell command run hone se pehle aapki permission lo. 3-tier safety model samajhna. Spec likho, agent code likhe.

🏗️ **PO SYSTEM ANALOGY**

James ki company: $500 se upar har Purchase Order pe manager ka sign chahiye. Kisi ne procurement access nahi khoya — sirf ek approval step add hua. Door khula, doorbell lagi. Yahi Tier 2 plugin hai!

## 3-Tier Safety Model:

| Tier | Mechanism | Enforcement | Kya Rokta Hai |
|---|---|---|---|
| 1 | Tool Profiles (coding/messaging) | In-process Node.js | Unauthorized tool access |
| 2 | Plugin Hooks (requireApproval) | In-process, plugin intercepts | Sensitive ops bina approval ke |
| 3 | NemoClaw Sandboxing | Out-of-process kernel-level | API key exfiltration, network escape |

## Approval Plugin Banwao (Agent Se):

```
Build me an OpenClaw plugin called "my-approval-gate" that requires
my approval on WhatsApp before any shell command runs.

Read the technical reference at: [is page ka URL]

After creating files in ~/.openclaw/plugins/my-approval-gate/, register:
- Set plugins.load.paths to include ~/.openclaw/plugins
- Enable plugins.entries.my-approval-gate
- Add approvals.plugin config (enabled: true, mode: "session")
- Restart gateway, run openclaw plugins list --verbose to confirm
```

⚠️ **Silent Plugin Failure!**

6 constraints hain jo plugins ko silently fail karte hain — code compile hota hai, plugin load hota hai, kuch nahi hota. Isliye URL dena zaroori hai taake agent technical constraints padhe aur sahi code likhe.

✅ **Lesson Ka Asli Siksha:**

James ne constraints likhe, code nahi. Agent ne code likha. "The actual skill: telling it what to build. Precisely enough that it does not get it wrong." — Spec writer bano, coder nahi.

# 14 Deploy to Production

VPS | Hetzner $5/month | 3 Deployment Paths

🎯 **IS LESSON MEIN KYA SIKHENGE**

Laptop se VPS pe migrate karna taake agent 24/7 run kare. 3 deployment paths. Hetzner $5/month pe complete setup.

James ne problem notice ki: "Jab laptop band karta hoon, agent 11:47 PM pe ruk jata hai." Solution: Cloud server pe deploy karo.

## 3 Deployment Paths:

| Path | Kiske Liye | Faida | Cost |
|---|---|---|---|
| **Managed** (Alibaba Cloud) | Beginners, zero CLI experience | Pre-installed image, no SSH, no firewall config | $0.99/month + free tokens |
| **VPS Native** (Hetzner) | SSH comfortable, hands-on learning | Same CLI commands as local — best for learning | ~$5/month |
| **VPS Docker** | Already running Docker Compose | Same isolation model | Same as VPS |

## Hetzner VPS Native (Recommended for Learners):

```
# Server: 2 vCPUs, 4GB RAM, 40GB SSD
# Time budget: 45 minutes
# - 15 min: VPS setup
# - 15 min: OpenClaw + onboarding
# - 15 min: "Paper cuts" (common issues)

# VPS pe same install command:
curl -fsSL https://openclaw.ai/install.sh | bash
openclaw onboard --install-daemon
```

⚠️ **Alibaba Cloud — Singapore Region Zaroori!**

Model Studio mein Singapore region select karo. Har model 1 million free tokens deta hai. Default region mein sab charged hai — Qwen Max bohot mahanga hai, 24 ghante mein account suspend ho sakta hai.

✅ **Deploy ke baad:**

Agent kabhi nahi sota, laptop close karne se kuch nahi hota, heartbeats 24/7 run karte hain, cron jobs exact time pe fire karte hain. Ye pehla "Production AI Employee" hai!

# 15 Isolate with NemoClaw

4-Pod Architecture | Privacy Router | API Key Vault | K3s

🎯 **IS LESSON MEIN KYA SIKHENGE**

Docker Compose ka security gap samajhna. NemoClaw ka 4-pod architecture. Privacy Router — API key agent tak pahunch nahi sakti. "Agent promises" vs "Agent physically cannot".

⚠️ **Docker Compose Security Gap:**

API key container environment variable mein hai. Agar attacker prompt inject kare aur `printenv` run karwaye → API key expose! Tool profiles in-process hain — same Node.js process enforce bhi karta hai aur use bhi karta hai.

## NemoClaw = OpenClaw + OpenShell + Privacy Router + Policy Engine

Ek install command → K3s cluster (lightweight Kubernetes) → 4 pods → out-of-process policy enforcement.

## 4-Pod Architecture:

| Pod | Kya Karta Hai | API Keys? | Agent? |
|---|---|---|---|
| **Gateway** | Channel adapters, auth, routing, Control UI | ❌ Nahi | ❌ Nahi |
| **Sandbox** | OpenClaw runtime, agent loop, tools, skills | ❌ Nahi | ✅ Haan |
| **Privacy Router** | API key vault, provider fallback, rate limiting | ✅ Haan | ❌ Nahi |
| **Policy Engine** | Landlock (fs), seccomp (syscalls), netns (network) | ❌ Nahi | ❌ Nahi |

🔑 **PRIVACY ROUTER — SABSE IMPORTANT DIFFERENCE**

Agent inference call karta hai `inference.local:8080` pe — koi API key nahi. Privacy Router alag pod mein hai, real API key add karta hai, Anthropic/OpenAI ko forward karta hai. Agent ko API key kabhi nahi milti — chahe root access ho!

## Prompt Injection Attack — NemoClaw vs Docker Compose:

| Attack | Docker Compose Result | NemoClaw Result |
|---|---|---|
| printenv | API key exposed! | No keys in env |

| | | |
|---|---|---|
| Config file read | Keys in config | Endpoint: inference.local, no credentials |
| Network sniff | Auth headers visible | No auth headers (router adds separately) |
| Direct API call | Possible | Network namespace sirf inference.local route karta hai |

✅ **Kab Upgrade Karein?**

Trust model change hone pe — scale pe nahi. Agar aap customers ke data touch kar rahe ho, agar untrusted content process ho raha hai, agar multi-tenant environment hai — tab NemoClaw. $5/month extra. Ek install command.

## 📚 Chapter 56 — Master Key Terms

| Term | Matlab | Lesson |
|---|---|---|
| Personal AI Employee | Always-on, multi-channel, extensible, owned AI worker | L1 |
| 6 Dimensions | Multi-channel, Always-on, Proactive, Extensible, Multi-agent, Ownership | L1 |
| Agent Loop | Intake → Context → Inference → Tool Execution → Reply | L3 |
| Tool Profile | Binary gate: coding/messaging/minimal/full — kya allowed hai | L3 |
| SOUL.md | Agent ka persona, tone, boundaries file | L4 |
| Emma's Rule | Agar instruction behavior nahi badalta — delete karo | L4 |
| MEMORY.md | Long-term curated memory — har session load hoti hai | L5 |
| Slash Commands | Zero tokens, gateway intercept — model bypass karte hain | L5 |
| Skill | SKILL.md folder — knowledge + workflow + scripts | L6 |
| ClawHub | Public skill registry — npm for agent capabilities | L6 |
| MCP Server | External tool access via protocol (stdio/SSE/streamable-http) | L7 |
| Silent Failure | MCP config galat → no error in chat, only in log | L7 |
| Heartbeat | Periodic self-check — har 30 min, HEARTBEAT.md se | L8 |
| HEARTBEAT_OK | Silent token — gateway message drop karta hai | L8 |
| TTS Modes | off / always / inbound / tagged | L9 |

| | | |
|---|---|---|
| Inbound Mode | Voice to voice, text to text — customer modality match | L9 |
| sessions_spawn | Subagent spawn karne ka tool (model quality pe depend) | L12 |
| 3-Tier Safety | Tool Profiles → Plugin Hooks → NemoClaw Sandboxing | L13 |
| Activation Dance | list → disabled → enable → configure (har feature) | L2, L9 |
| Privacy Router | API key vault — agent is key ko kabhi nahi dekh sakta | L15 |
| 4-Pod Architecture | Gateway + Sandbox + Privacy Router + Policy Engine | L15 |
| gog CLI | Google Workspace bridge — Gmail, Calendar, Drive | L11 |
| Gateway Config | Sab agents ke liye shared (plugins, TTS, model) | L10 |
| Workspace Config | Per-agent individual (SOUL.md, skills, MCP) | L10 |

Prepared for: **Imran Hassan** | 2026

Source: agentfactory.panaversity.org | Chapter 56 — Meet Your Personal AI Employee

Print PDF: Ctrl+P → Save as PDF